

CodeCraft Challenges

Problems with their objectives (Mobile App Development Challenges)

SR No.	Problem	Objectives (What the chatbot could do)
1	University Event Management	• Display upcoming seminars, workshops, cultural events, and competitions. • Students can register through the app. • Organizers can manage participants and attendance.
2	Campus Bus Tracking	• Students see university bus routes and schedules. • Live bus location using GPS. • Notifications about delays or route changes.
3	University Emergency & Safety App	• One-tap emergency contacts. • Campus security information. • Report unsafe situations. • Share location during an emergency
4	Placement Cell Mobile Application	• Centralize all placement-related information. • Notify eligible students about new job opportunities. • Allow students to register for placement drives. • Automatically check eligibility based on CGPA, branch, backlogs, etc. • Track student applications and selection status.
5	Research Cell Mobile Application	• Centralize university research information. • Maintain researcher and faculty profiles. • Manage research projects and publications. • Notify users about funding and conference opportunities. • Track project proposals and approval status. • Maintain records of patents, papers, books, and conferences.

Competition Rules (Applicable Every Month)

- Individual or team (maximum 2 students)
- Time limit: 2–3 hours
- Original work only
- GitHub repository submission (optional but recommended)
- 5–10-minute project demonstration
- Source code and documentation must be submitted before the deadline

Suggested Technologies

- **Languages:** Python, Java, JavaScript
- **Frontend:** HTML, CSS, Bootstrap, React
- **Backend:** Flask, Django, Node.js
- **Database:** SQLite, MySQL, MongoDB
- **AI/NLP:** NLTK, spaCy, Hugging Face Transformers, OpenAI API, Gemini API
- **Deployment:** Streamlit, Render, Vercel, Railway